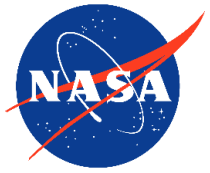


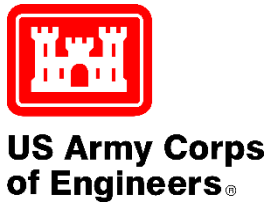
Post-processing and Data Visualization

Everyone

Data Visualization Tools

- Write Python (or other language) viewer to visualize the data, generate plots, etc.
- Panoply netCDF, HDF, and GRIB data viewer
  https://www.giss.nasa.gov/tools/panoply/
- GRADS
  - http://cola.gmu.edu/grads/
- NCView
  - Generally native NetCDF data viewer on linux
- Integrated Data Viewer
  - https://www.unidata.ucar.edu/software/idv/
- View NetCDF data in GIS systems (QGIS)
  - LIS output has to be converted to GeoTIFF first using GDAL tools











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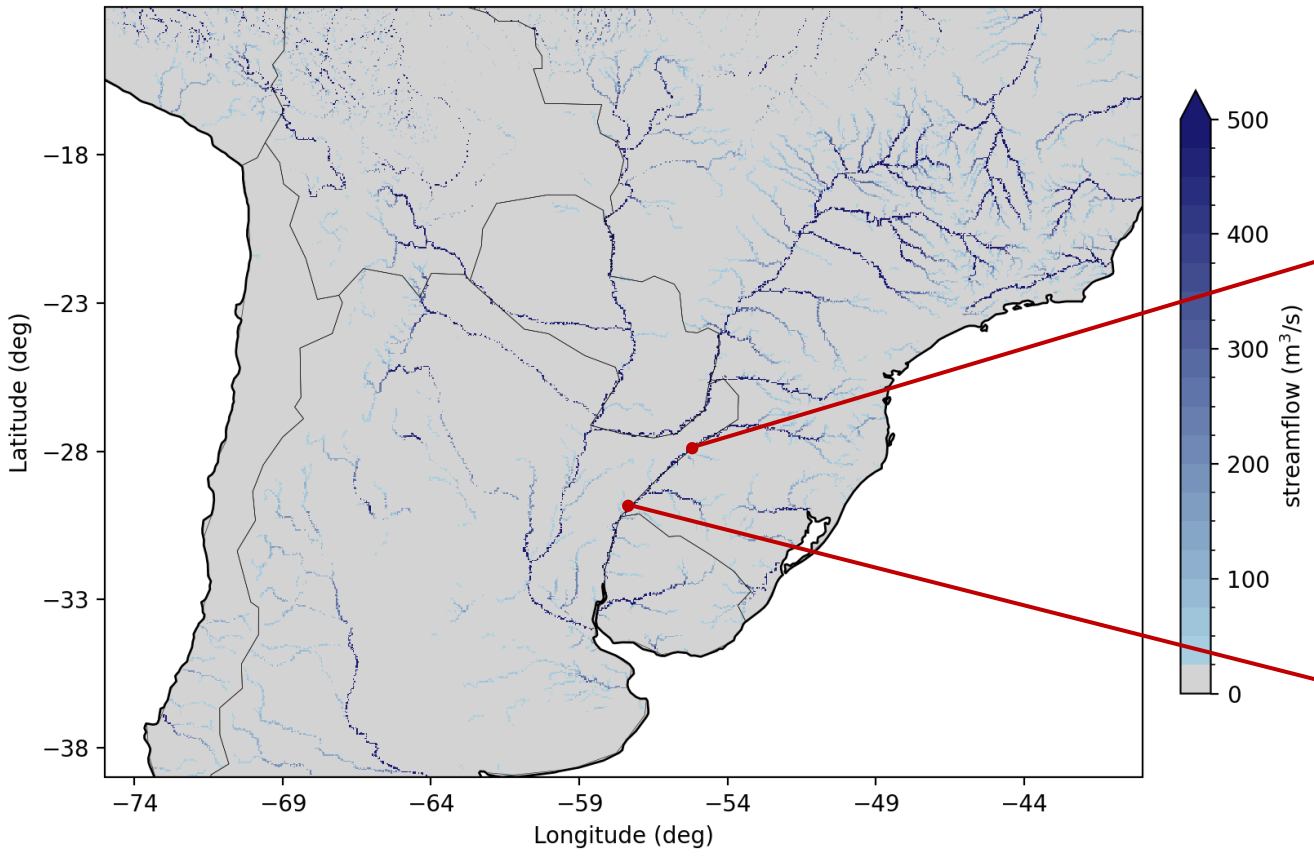
US Army Corps of Engineers

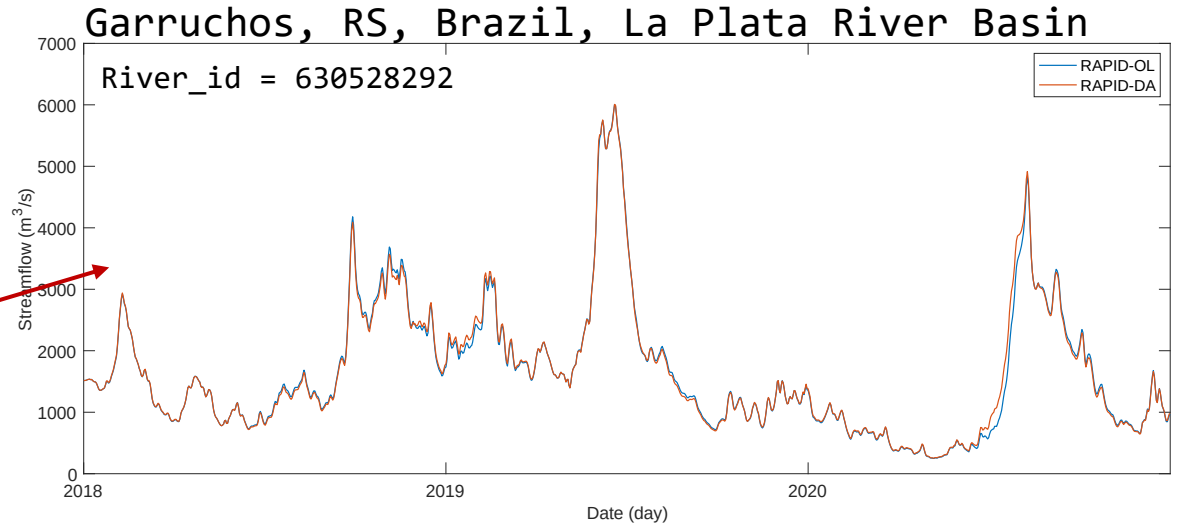
ERDC

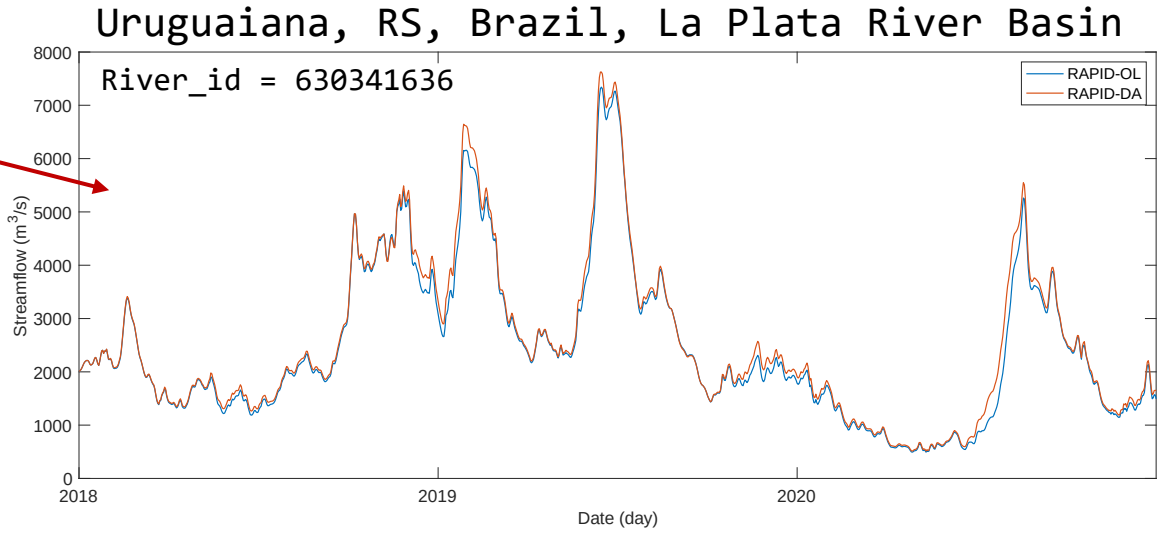
IWR

University of Virginia

Streamflow from RAPID







RAPID DA: Jan 01, 2020

Python 3 visualization

Python 3 codes

```
def main(input_file, output_file):
    # load the location file for converting 1D to 2D format.
    # Locations were calculated using the nearest neighborhood
    # method between river_id and LIS grid.
    mat = sio.loadmat('loc2d.mat')
    idx=np.array(mat['idx']).transpose()
    idy=np.array(mat['idy']).transpose()

    # read LIS-RAPID netcdf file (1D format)
    dataset = nc.Dataset(input_file, 'r')
    q_rapid1d = dataset.variables['Qout'][:]
    dataset.close()

    # laplata basin
    lat=np.linspace(-38.98,-13.02,650)
    lon=np.linspace(-74.98,-40.02,875)

    # Convert 1D to 2D
    q_rapid2d = np.ones((len(lon),len(lat))) * -9999.0
    for i in range(len(idx)):
        q_rapid2d[idx[i]-1, idy[i]-1] = q_rapid1d[i]

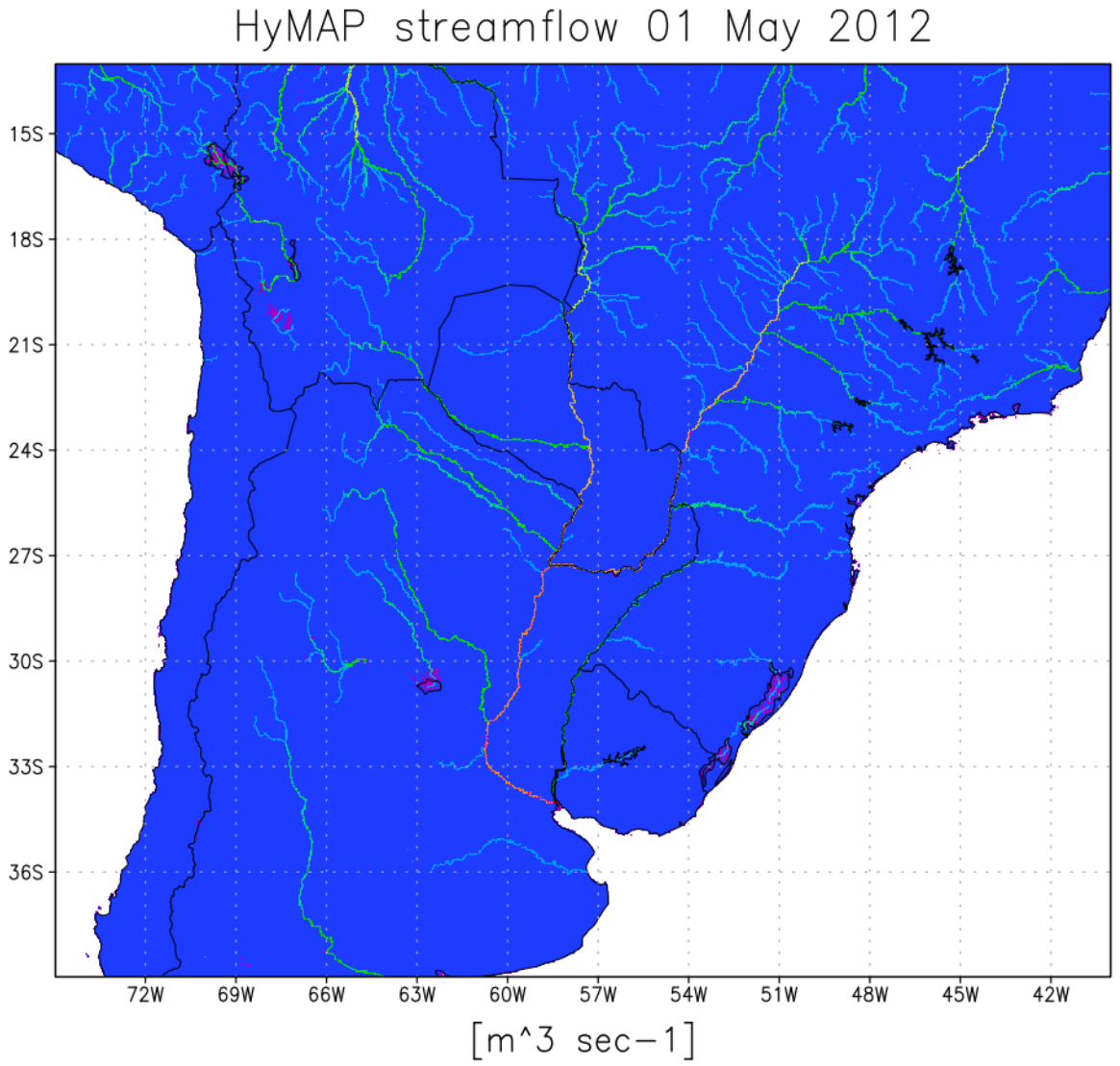
    q_rapid2d_t=np.transpose(q_rapid2d)
```

```
# Set Dataset for plot
ds = xr.Dataset({
     'q_rapid2d': xr.DataArray(
                  data   = q_rapid2d_t,
                  dims = ['latitude','longitude'],
                  coords = {'latitude': lat, 'longitude': lon},
                  attrs  = {
                      '_FillValue': -999.9,
                      'units'     : 'm3/s'}
                  )
              },
          attrs = {'attr': 'RAPID attribute'}
    )
 …

 # Generate figure
 U = ds.q_rapid2d
 q = U.plot.imshow(ax=ax, transform=projection, cmap=cmap,
     norm=norm, add_colorbar=False)
 …
```

> ./review_rapid.py input_file (e.g., LIS_HIST_202001010000.d01.nc) output_file
Note that generated plot lacks scientific significance.

Streamflow from HyMAP



This figure was generated with:

GrADS

http://cola.gmu.edu/grads/

GrADS is free software that can quickly make nice figures from gridded datasets.

One nice feature of GrADS is that it can “template” in time, so a user can plot results from many different files in a single figure.

GrADS visualization

```
DSET   ^LIS_HIST_201205010000.d01.nc
DTYPE   netcdf
UNDEF   -9999.
XDEF east_west     875 LINEAR   -74.98  0.04
YDEF north_south   650 LINEAR   -38.98  0.04
```

We write a short ASCII/text file for GrADS to open the dataset:

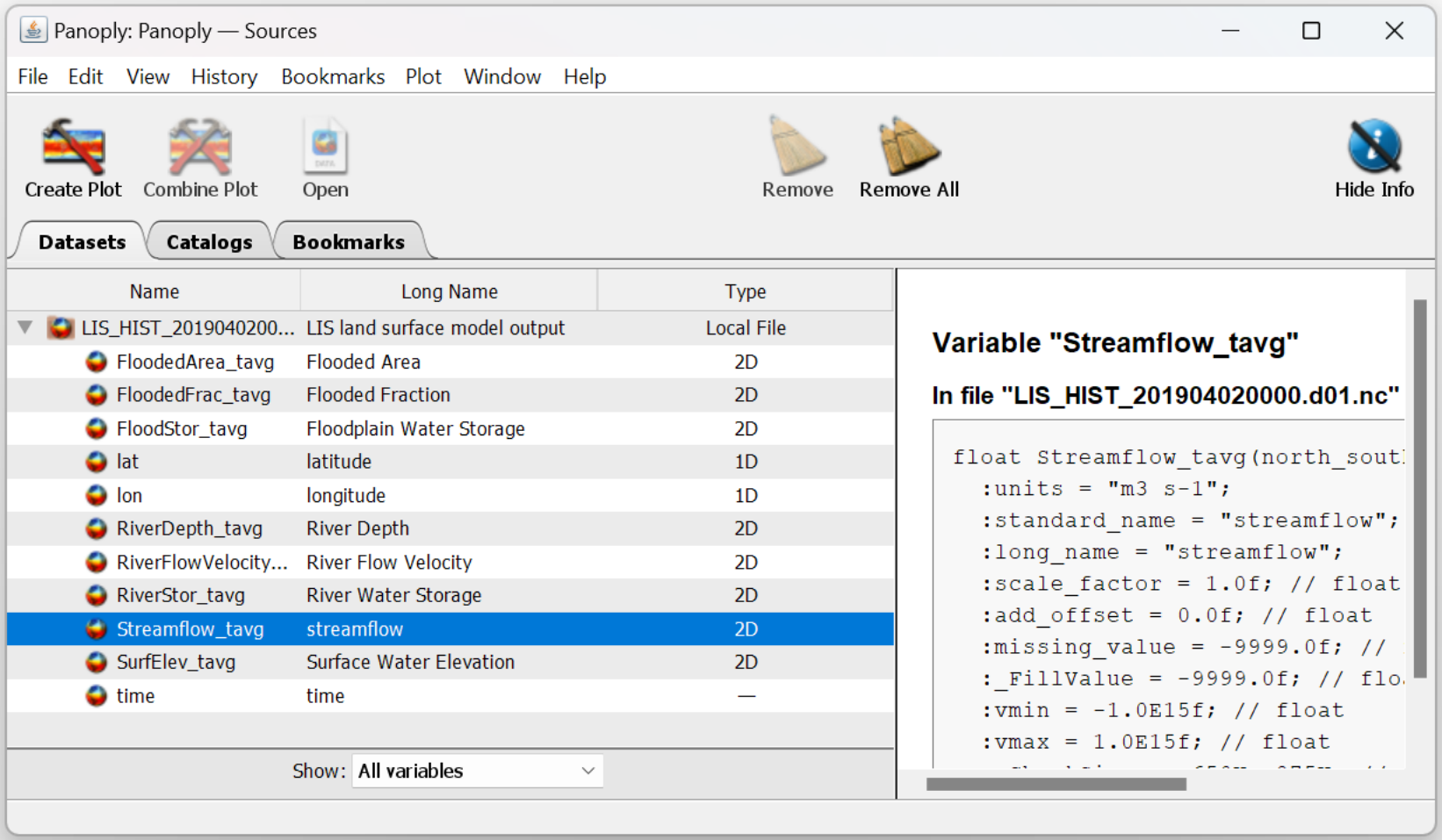
1) Point to the data file
2) The data file type is netCDF
3) Set the “undefined” value
4) Set our domain (875 by 650 grid points, 0.04-degree grid spacing, with the center of the lower-left grid point at 38.98 South and 74.98 West)

To generate a plot similar to the one on the previous page, do this in GrADS:

```
xdfopen [name-of-short-text-file]
set gxout grfill
set clevs 0 100 500 1000 5000
1000
          20000 50000
d streamflow
run cbarn.gs
draw title HyMAP Streamflow
draw xlab [m^3 sec-1]
gxprint plot-name.png
```

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Panoply


Panoply: Panoply — Sources
File Edit View History Bookmarks Plot Window Help
Create Plot
Combine Plot
Open
Remove
Remove All
Hide Info
Datasets
Catalogs
Bookmarks
Name
Long Name
Type
LIS_HIST_2019040200...
LIS land surface model output
Local File
FloodedArea_tavg
Flooded Area
2D
FloodedFrac_tavg
Flooded Fraction
2D
FloodStor_tavg
Floodplain Water Storage
2D
lat
latitude
1D
lon
longitude
1D
RiverDepth_tavg
River Depth
2D
RiverFlowVelocity...
River Flow Velocity
2D
RiverStor_tavg
River Water Storage
2D
Streamflow_tavg
streamflow
2D
SurfElev_tavg
Surface Water Elevation
2D
time
time
—
Show: All variables
Variable "Streamflow_tavg"
In file "LIS_HIST_201904020000.d01.nc"
float Streamflow_tavg(north_sout
:units = "m3 s-1";
:standard_name = "streamflow";
:long_name = "streamflow";
:scale_factor = 1.0f; // float
:add_offset = 0.0f; // float
:missing_value = -9999.0f; //
:_FillValue = -9999.0f; // flo
:vmin = -1.0E15f; // float
:vmax = 1.0E15f; // float

Convert LIS data to GeoTIFF for GIS

```
INFILE=${1}
PARAM=${2}

LRLAT=-90.0
ULLON=0
ULLAT=90.0
LRLON=360.0

gdal_translate NETCDF:"${INFILE}":${PARAM} -of GTiff -a_ullr ${ULLON} ${ULLAT} ${LRLON} ${LRLAT} -a_srs EPSG:4326 --config
CENTER_LONG 0 "${INFILE}.${PARAM}.tif"

gdalwarp -t_srs WGS84  "${INFILE}.${PARAM}.tif" "${INFILE}.${PARAM}.translated.tif" --config CENTER_LONG 0
```

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